

Rating Gaps and Effective Resistance: Exact Fluctuation Laws for Elo and Bradley–Terry Systems

Cheng Qian*

July 2026

Abstract

A rating system is a network: players are nodes, matches are edges, and ratings are node potentials whose differences predict outcomes. We show that the fluctuation behavior of rating *gaps* is governed by a single question — *where noise enters the network relative to where matches defend it* — and that the answer splits rating systems into three regimes with three different laws. (i) *Estimation*: the variance of the maximum-likelihood estimate of any rating gap equals the effective resistance between the two players in the match graph, with Fisher conductances. (ii) *Dynamics with static skills*: the stationary gap fluctuation of online Elo is K (the update factor) for *every* pair on *every* connected graph — outcome noise and restoring force enter through the same edges and their graph dependence cancels exactly. (iii) *Dynamics with drifting skills*: idiosyncratic skill drift restores the resistance law as the tracking error, with mean reversion acting as a grounding conductance on every node. All three laws are verified in simulation; the estimation law is additionally confirmed on up to 104,419 real Chatbot Arena battles with a zero-parameter fit (slope 1.035–1.041 against the theoretical 1, $R^2 \geq 0.958$), in both dense and clustered battle graphs. The clustered experiment — two collection eras joined by five bridge models — confirms the framework’s distinctive prediction: the resistance law stays calibrated where the battle-count heuristic drifts by 17%, and quantifies how the comparability of 103 models across a year of LLM history rests on five survivors. Applications include resistance-based leaderboard confidence intervals, a matchmaking design rule, and a structural explanation of chess’s era problem.

1 Introduction

Pairwise-comparison rating systems — Elo in chess [3], its Bradley–Terry foundation [1, 2], Glicko [4], TrueSkill [5], and most recently LLM leaderboards such as Chatbot Arena [9] — share one structure: a set of entities, a graph of realized comparisons, and a vector of ratings whose pairwise *differences* are the only predictive content. The absolute level of a rating is pure convention (the familiar incomparability of ratings across pools); the gaps are the physics.

This paper studies the fluctuations of those gaps and shows they are organized by a single structural question:

Where does randomness enter the comparison network, relative to where the matches act?

*Code, data pipelines, and the companion monograph are available at <https://github.com/macrokitt/fixed-gaps>. This paper develops Chapter 7 of the monograph.

Three answers give three laws. Randomness in *outcomes* enters on the edges — the same edges through which every match pulls the rating gap toward the observed result. Randomness in *true skills* (players improving and declining) enters at the nodes. And randomness *common* to all players enters nowhere: it cancels from every gap identically.

Contributions.

1. **Estimation resistance law** (Thm. 1). The variance of the maximum-likelihood estimate of any rating gap — between any two players, whether or not they ever met — equals the effective resistance between them in the match graph with Fisher conductances. The structurally closest prior work is Boumal et al. [10], who derive Cramér-Rao bounds for rotation synchronization as the pseudoinverse of a weighted graph Laplacian — the same identification; their setting is $SE(d)/SO(d)$ -valued signals, not rating scalars. Röttger et al. [11] connect optimal design of Bradley–Terry experiments to graph Laplacians but do not state the per-pair variance formula. The explicit identification $\text{Var}(\hat{r}_i - \hat{r}_j) = R_{\text{eff}}(i, j)$ in the rating setting, and the consequences drawn from it, appear to be new.
2. **Isotropy of online Elo** (Thm. 2). With static skills, the stationary fluctuation of the rating gap under online Elo with update factor K is $\text{Var}(r_i - r_j) = K$ for every pair, on every connected match graph: a path and a complete graph produce identical long-run churn. To our knowledge this graph-independence is unobserved in the literature; it is invisible unless one asks the noise-location question, since it results from an exact cancellation between edge noise and edge defense.
3. **Tracking resistance law** (Thm. 3). When true skills drift idiosyncratically (variance rate σ_d^2 per node), the resistance law returns as the stationary *tracking* error: $\text{Var}((r_i - r_j) - (s_i - s_j)) = K + \frac{2|E|\sigma_d^2}{K} R_{\text{eff}}(i, j)$, and mean reversion of skills at rate θ replaces R_{eff} by the resistance of the network in which every node is grounded through conductance θ/a . Without mean reversion, the logistic saturates and no bounded-update system can track at all (Prop. 1).
4. **Validation.** All laws verified in controlled simulation (§4); the estimation law verified on real Chatbot Arena data with no free parameters, in a dense regime (39,241 battles, 60 models, slope 1.035, $R^2 = 0.958$) and in a clustered two-era regime (104,419 battles, 108 models, slope 1.041, $R^2 = 0.963$), where the framework’s distinctive prediction — calibration of the resistance prediction where the count heuristic fails — is confirmed (§5).
5. **Design consequences** (§6): resistance-based confidence intervals for leaderboards; a matchmaking rule (bridge comparisons dominate: one cross-cluster match outperformed a within-cluster match by a factor ~ 44 on worst-pair resistance in our test configuration); and a derivation of chess’s era-comparison problem as a chain instability, together with an explanation of why Arena’s era problem is mild.

2 Setup

Players $1, \dots, n$ carry true skills $s \in \mathbb{R}^n$; player i beats j with probability $\sigma(s_i - s_j)$, $\sigma(x) = (1 + e^{-x})^{-1}$ (Bradley–Terry; Elo’s base-10/400 scale is a linear change of units). The *match*

graph $G = (V, E)$ records which pairs play; L denotes its Laplacian and, for conductances c_e on the edges, L_c the weighted Laplacian. The *effective resistance* between i and j is $R_{\text{eff}}(i, j) = (e_i - e_j)^\top L_c^+ (e_i - e_j)$ with L_c^+ the pseudoinverse [6]. Online Elo updates, after a match on edge (i, j) with score $x \in \{0, 1\}$:

$$r_i += K(x - \sigma(r_i - r_j)), \quad r_j -= K(x - \sigma(r_i - r_j)),$$

which is stochastic gradient ascent on the Bradley–Terry log-likelihood [8]. Matches arrive uniformly on E ; linearizing about small gaps ($\sigma'(0) = \frac{1}{4}$) gives the drift–diffusion

$$dr = a L (s - r) dt + dM, \quad a = \frac{K}{4|E|}, \quad \text{Cov}(dM) = \frac{K^2}{4|E|} L dt. \quad (1)$$

The structural fact driving everything below: *the noise covariance and the restoring force are built from the same Laplacian*, because in a rating system every observation of a gap is also a defense of it.

3 Three laws

Theorem 1 (Estimation resistance law). *With n_e matches on edge e and static skills, the Fisher information of the Bradley–Terry model is L_c with $c_e = n_e p_e(1 - p_e)$, $p_e = \sigma(s_i - s_j)$. Hence, asymptotically, for every pair (i, j) ,*

$$\text{Var}(\widehat{r_i - r_j}) = R_{\text{eff}}(i, j) \quad \text{in the match graph with Fisher conductances;}$$

at equal skills and m matches per edge this is $(4/m) R_{\text{eff}}^{\text{unit}}(i, j)$.

Proof. The log-likelihood Hessian at the truth is $-\sum_e n_e p_e(1 - p_e)(e_i - e_j)(e_i - e_j)^\top = -L_c$; the MLE covariance on the mean-zero gauge is L_c^+ , and the variance of the contrast $(e_i - e_j)^\top \hat{r}$ is the stated quadratic form, i.e. the resistance [7]. $\square$

Uncertainty about a gap is thus a property of the *network between* the pair — parallel comparison paths add like parallel conductances — not of the two players’ match counts.

Theorem 2 (Isotropy of stationary Elo). *Under (1) with static equal skills, the stationary covariance of r on the mean-zero subspace is $\Sigma = \frac{K}{2} P$ (with P the centering projector). Consequently*

$$\text{Var}(r_i - r_j) = K \quad \text{for every pair, on every connected match graph.}$$

Proof. The stationary Lyapunov equation for $dr = -aLr dt + dM$ is $aL\Sigma + \Sigma La = \frac{K^2}{4|E|} L$. Since drift and noise are simultaneously diagonalizable (both are L), $\Sigma = \frac{K^2/(4|E|)}{2a} P = \frac{K}{2} P$ on the range of L , and $(e_i - e_j)^\top P(e_i - e_j) = 2$. $\square$

The graph dependence cancels exactly: a poorly connected pair receives less restoring force but, in the same proportion, less noise. Online Elo’s long-run churn is graph-independent — contrary to the natural intuition that remote pairs should wander more, and, to our knowledge, unremarked in the rating literature.

Theorem 3 (Tracking resistance law). *Let true skills drift idiosyncratically, $ds = \sigma_d dW$ (independent across players), and let $e = r - s$ be the tracking error under (1). Then in stationarity*

$$\text{Var}(e_i - e_j) = \underbrace{K}_{\text{edge noise (Thm. 2)}} + \underbrace{\frac{2|E|\sigma_d^2}{K} R_{\text{eff}}(i, j)}_{\text{node noise: the resistance law}}.$$

If instead skills are mean-reverting, $ds = -\theta s dt + \sigma_d dW$, the exact linear theory replaces R_{eff} by the grounded resistance — the quadratic form of $(L + (\theta/a)I)^{-1}$: mean reversion is a grounding wire of conductance θ/a on every node.

Proof. Mode-by-mode on the eigenbasis of L (eigenvalue λ): the error obeys $de_\lambda = -a\lambda e_\lambda dt + (\text{edge noise, rate } \frac{K^2}{4|E|}\lambda) - (\text{node noise, rate } \sigma_d^2)$, giving stationary variance $\frac{K}{2} + \frac{\sigma_d^2}{2a\lambda}$; summing λ^{-1} against the pair contrast yields R_{eff} . With mean reversion the transfer function from skill noise to error acquires the factor $-i\omega/((i\omega + \theta)(i\omega + a\lambda))$, whose integrated power replaces $(a\lambda)^{-1}$ by $(\theta + a\lambda)^{-1}$, i.e. grounds the network. $\square$

Proposition 1 (Saturation: the era catastrophe). *If skills random-walk without mean reversion, skill gaps grow without bound, $\sigma'(s_i - s_j) \rightarrow 0$, and the linearization gain in (1) collapses: the tracking error of a bounded-update system diverges on every graph.*

Common shocks — randomness applied to all players alike — cancel from every gap identically and are invisible to all three laws: only *asymmetric* randomness moves gaps. The trichotomy common/node/edge is exhaustive and is the organizing principle of this paper.

4 Simulation

We verify the laws against the *actual* (nonlinear, discrete) Elo dynamics, not the linearization. Configurations: $n = 12$ – 16 players on path, cycle, two-clique-plus-two-bridges, random 3-regular, and complete graphs; $K = 0.1$. Code and committed output accompany the paper.

Thm. 1: regressing empirical MLE gap-error variance (600 trials, 40 matches/edge) on $(4/m)R_{\text{eff}}$ over all pairs gives slope 1.006–1.027 (theory 1) with R^2 up to 0.994 across the graph families (the complete graph has all resistances nearly equal, hence no variance to explain).

Thm. 2: online Elo, 4×10^6 steps, static skills: mean stationary gap variance 0.1055–0.1060 against the predicted $K = 0.1$ (the small excess is the expected nonlinear correction), and flat in resistance — on the path, the low-resistance half of pairs shows 0.1056 and the high-resistance half 0.1064.

Thm. 3: drifting skills ($\sigma_d = 0.004$, $\theta = 10^{-4}$, 8×10^6 steps): tracking-error variance is linear in R_{eff} (path $R^2 = 0.95$, two-cluster 0.90), intercept 0.106/0.105 vs. the predicted $K = 0.1$; measured slopes 0.00206 (path) and 0.00964 (two-cluster) against exact grounded-theory values 0.00213 (3%) and 0.00812 (within 20%), with the naive ungrounded value (0.00352 on the path) clearly rejected.

Prop. 1: with $\theta = 0$ the path-graph tracking variance reaches $\sim 60\times$ the defended level and keeps growing.

5 Real data: Chatbot Arena

5.1 Dense regime

From the public 55k Arena preference dataset we retain decisive battles among models with ≥ 200 of them: 39,241 battles, 60 models, 1,145 of 1,770 pairs realized (density 0.65). We fit the Bradley–Terry MLE, compute the predicted variance of every pairwise gap as the Fisher resistance (Thm. 1), and compare against 200 bootstrap resamples of the battles. The theory has no free parameters: the predicted slope is exactly 1. Over all 1,770 pairs:

$$\text{slope} = 1.035, \quad \text{corr} = 0.979, \quad R^2_{\text{through origin}} = 0.958.$$

The law is quantitatively right at the few-percent level, simultaneously across every pair, including the majority that never met. On this dense graph the folk heuristic $1/n_i + 1/n_j$ also correlates well (log–log 0.956 vs. 0.987 for resistance): in dense graphs the resistance approximately *reduces* to the count formula — the theory explains when the heuristic works and predicts where it fails, namely clustered graphs. Which motivates:

5.2 Clustered regime: the era experiment

Merging the 55k collection (2023–early 2024) with the 100k collection (June–August 2024) yields a two-era graph: 104,419 decisive battles, 108 models — 55 exclusive to era 1, 48 exclusive to era 2, and a **five-model bridge** (three GPT-4 variants, GPT-3.5-turbo, Mixtral) through which all $55 \times 48 = 2,640$ cross-era comparisons must route, since era-exclusive models never coexisted. Findings:

1. The resistance law holds on the clustered graph: slope 1.041, corr 0.982, $R^2 = 0.963$ over all 5,778 pairs.
2. Calibrating the count heuristic on within-era pairs and evaluating the median ratio of bootstrap variance to prediction: the resistance prediction reads 1.01 within-era and 1.01 cross-era — identically calibrated in both regimes with no adjustment — while the count heuristic reads 0.92 within vs. 1.08 cross, a systematic 17% spread in the predicted direction.
3. The bridge is load-bearing: deleting a single bridge model’s battles raises median cross-era resistance by up to 8%; deleting all five disconnects the eras — cross-era gaps become infinitely uncertain. The comparability of 103 models across a year of LLM history rests, in this precise sense, on five survivors.
4. Arena’s cross-era penalty is nonetheless mild ($\sim 8\%$ in variance at matched counts), because its bridge models are battle-rich: the platform keeps old flagships in rotation, which is exactly the long-range-edge prescription of §6. Chess, whose bridges (individual careers spanning eras) are thin and unreplayable, is the same graph with weak bridges — hence its severe era problem.

6 Design consequences

Resistance-based confidence intervals. By Thm. 1, leaderboard uncertainty for “ A vs. B ” should be reported as the resistance between A and B in the battle graph. Two models with

thousands of battles each but no common opponents can have arbitrarily uncertain relative rank; conversely a lightly played model wired into a well-connected neighborhood can be sharply placed.

Matchmaking. To sharpen a leaderboard fastest, schedule the comparison that most reduces the (grounded) resistance of the pairs of interest. In our two-cluster test, one bridge match reduced worst-pair resistance from 1.000 to 0.778; one more within-cluster match, to 0.995 — a $\sim 44\times$ difference per match. Platforms that matchmake for entertainment systematically overspend on low-resistance edges.

The era problem. The temporal comparison graph of a long-lived rating pool is chain-like; by Thm. 3 cross-era gaps drift with variance linear in era separation (and by Prop. 1, without anchoring, tracking eventually fails outright). Known remedies map onto the theory: anchoring to a fixed reference is grounding; careers spanning eras are long-range edges; Arena’s bridge-rich rotation policy is the design rule implemented.

7 Related work

The Bradley–Terry model and its MLE date to [1, 2]; Elo’s system to [3]; measurement-error refinements include Glicko [4] and TrueSkill [5]. Chatbot Arena and its Bradley–Terry leaderboard are described in [9], which uses bootstrap confidence intervals; our Thm. 1 gives those intervals a closed structural form, and §5 verifies it. For calibrated Elo confidence intervals via conformal prediction, see [13]; [14] studies noise calibration and admissible ranking regions in nonreciprocal pairwise comparisons.

Effective resistance is classical in the random-walk literature [6] and minimizing it is studied in [7]. The structurally closest prior work to Law (i) is Boumal et al. [10], who show that the Cramér–Rao bound for rotation *synchronization* equals the pseudoinverse of a weighted graph Laplacian — the same structure; their setting is $SE(d)/SO(d)$ -valued signals, not rating scalars. Röttger et al. [11] connect optimal design of Bradley–Terry comparisons to graph Laplacians via the matrix-tree theorem but do not state the per-pair variance formula or the resistance interpretation. To our knowledge, the identification $\text{Var}(\hat{r}_i - \hat{r}_j) = R_{\text{eff}}(i, j)$ in the rating setting, and the consequences drawn from it, are new.

For Law (ii), Olesker-Taylor and Zanetti [12] analyze stationary Elo via Markov chains and provide spectral-gap-dependent upper bounds on average gap variance; our isotropy result (Thm. 2) is strictly sharper — an exact formula showing the spectral dependence cancels entirely. The interpretation of SGD’s stationary distribution as an Ornstein–Uhlenbeck process follows [8]. We are not aware of prior statements of the grounded-resistance tracking law (Thm. 3) in this setting. The general framework relating noise location, defended invariants, and effective resistance is developed in the companion monograph (footnote, p. 1), of which this paper is the self-contained applied chapter.

8 Limitations and outlook

The dynamical results are linearizations, and the simulations quantify the nonlinear corrections at the few-percent level for realistic K ; Thm. 2’s constant, in particular, is K only to leading

order. The Arena validation tests the estimation law; catching the tracking law (Thm. 3) in the wild requires a longitudinal pool with thin bridges — historical chess federation data is the natural target, but game-level archives at the required scale are not freely available. The era experiment treats the two collections as exchangeable judge populations; drift in judging standards would appear as a common shock and cancel from gaps, but interactions between judge mix and model identity would not, and bounding them is future work.

References

- [1] E. Zermelo. Die Berechnung der Turnier-Ergebnisse als ein Maximumproblem der Wahrscheinlichkeitsrechnung. *Math. Z.*, 29:436–460, 1929.
- [2] R. A. Bradley and M. E. Terry. Rank analysis of incomplete block designs: I. The method of paired comparisons. *Biometrika*, 39:324–345, 1952.
- [3] A. E. Elo. *The Rating of Chessplayers, Past and Present*. Arco, 1978.
- [4] M. E. Glickman. Parameter estimation in large dynamic paired comparison experiments. *J. R. Stat. Soc. C*, 48:377–394, 1999.
- [5] R. Herbrich, T. Minka, and T. Graepel. TrueSkill: a Bayesian skill rating system. *NeurIPS*, 2006.
- [6] P. G. Doyle and J. L. Snell. *Random Walks and Electric Networks*. MAA, 1984.
- [7] A. Ghosh, S. Boyd, and A. Saberi. Minimizing effective resistance of a graph. *SIAM Review*, 50(1):37–66, 2008.
- [8] S. Mandt, M. D. Hoffman, and D. M. Blei. Stochastic gradient descent as approximate Bayesian inference. *JMLR*, 18(134):1–35, 2017.
- [9] W.-L. Chiang et al. Chatbot Arena: an open platform for evaluating LLMs by human preference. *ICML*, 2024.
- [10] N. Boumal, A. Singer, P.-A. Absil, and V. D. Blondel. Cramér-Rao bounds for synchronization of rotations. *Information and Inference*, 3(1):1–39, 2014.
- [11] F. Röttger, T. Kahle, and R. Schwabe. Optimal designs for discrete choice models via graph Laplacians. *J. Stat. Theory Pract.*, 2022.
- [12] S. Olesker-Taylor and L. Zanetti. An analysis of Elo rating systems via Markov chains. *NeurIPS*, 2024.
- [13] B. Kargi and D. Salinas. From uncertain judgments to calibrated rankings: conformal Elo estimation for LLM evaluation. arXiv:2606.13221, 2026.
- [14] J.-P. Magnot. Noisy nonreciprocal pairwise comparisons: scale variation, noise calibration, and admissible ranking regions. arXiv:2604.04588, 2026.